

Ditshwanelo Tumane

Software Engineer | Cloud, AI & Full-Stack Specialist

tshwanelotumane2@gmail.com | Cape Town, South Africa | LinkedIn

PROFILE SUMMARY

Dynamic, multi-disciplinary Software Engineer with a proven and rapidly accelerating career trajectory spanning AI integration, cloud architecture, full-stack development, financial systems, and DevOps engineering. Uniquely positioned at the intersection of intelligent automation and practical software delivery — combining hands-on Azure AI Language and agentic AI expertise with production-grade full-stack engineering across Python, JavaScript (React.js/TypeScript), advanced SQL, and RESTful API architecture. Brings end-to-end technical range from cloud infrastructure management (AWS + Azure) and CI/CD pipeline operations to relational database architecture (PostgreSQL, 3NF) and client-facing application delivery. Currently embedded as Full Stack Developer at Lesaka Technologies — a JSE/NASDAQ-listed fintech leader — building financial services applications within a high-volume payment and merchant ecosystem, gaining direct exposure to financial data pipelines, billing-adjacent workflows, and API-driven cross-system integration at scale. Certified across Microsoft Azure AI Fundamentals, AWS Cloud Practitioner, and 25+ Azure specialisation badges. Previously selected as AI and Cloud Technical Lead for the Mandela Legacy Foundation through a competitive outsourced placement — a direct validation of above-average technical authority. Passionate about delivering high-performance, scalable solutions that bridge AI capability, cloud infrastructure, and real-world operational outcomes.

SKILLS

AI & Machine Learning: Azure AI Language Services | LLM Integration & Orchestration | Agentic AI Systems | Prompt Engineering | AI Agent Design | Intelligent Workflow Automation | Responsible AI | Model Evaluation

Cloud Platforms: Azure (AI Services, Cognitive Services, Functions, API Management, PostgreSQL, DevOps) | AWS (EC2, S3, Lambda, API Gateway, RDS) | Cloud Automation | Microservices

Programming & Full-Stack: Python | SQL | JavaScript | TypeScript | React.js | HTML5 | CSS3 | PHP | AJAX | REST API Development

Backend & Architecture: API-Driven Architecture | System Integration Patterns | Microservices | CI/CD Pipelines | Middleware | Distributed Systems Principles | Version Control (Git)

Data & Databases: PostgreSQL | Advanced SQL (DDL/DML) | Database Design (ERD, 3NF) | Data Modelling | Data Warehousing | Financial Data Pipelines | ETL Design | Data Validation

Professional: Technical Leadership | Architecture Decision-Making | Agile/Scrum | Stakeholder Communication | Cross-Functional Collaboration | Problem Solving | Project Management | SDLC

EMPLOYMENT HISTORY

Full Stack Developer

Lesaka Technologies | Cape Town, South Africa (Hybrid) *March 2026 – Present (Currently Working)*

- **Financial Application Development:** Develop and deploy full-stack financial services applications within Lesaka Technologies' consumer and merchant payment ecosystem — one of South Africa's leading JSE/NASDAQ-listed fintech companies — integrating intelligent data workflows and API-driven integration patterns to deliver scalable, high-availability financial services at transactional scale.
- **RESTful API & Integration Architecture:** Design and maintain RESTful APIs enabling real-time structured data exchange across consumer-facing and merchant payment platforms, applying API-driven integration patterns, idempotency design, and data validation logic consistent with enterprise backend architecture standards.

- **Financial Data Pipeline & Database Engineering:** Design and optimise PostgreSQL database schemas for high-volume transactional financial data, implement automated data validation and reconciliation logic, and build Python-driven data pipelines ensuring billing data accuracy and cross-system financial data integrity.
- **Cloud Infrastructure & CI/CD:** Deploy and manage cloud-hosted applications on AWS and Azure, implementing automated CI/CD pipelines, observability monitoring, and multi-environment deployment practices for mission-critical financial services delivery.
- **Cross-Functional Technical Collaboration:** Collaborate with Finance, Product, and Engineering teams to translate complex business requirements into scalable technical architectures and data-driven solutions, serving as a technical bridge between engineering delivery and business outcomes.

Software Engineer

Netcampus Group (Kaagisano) | Cape Town, South Africa (Hybrid) *March 2025 – March 2026*

- **AI Integration & LLM-Powered Application Design:** Designed and integrated Azure AI Language services — Microsoft's enterprise LLM cognitive platform — with PostgreSQL databases to architect intelligent, NLP-driven data applications, applying prompt engineering, model evaluation, and responsible AI governance in production cloud environments.
- **Agentic AI System Architecture:** Proactively applied advanced agentic AI knowledge to design AI agent workflows for document intelligence, information retrieval automation, and intelligent decision pipelines — demonstrating architectural depth and initiative well beyond foundational AI exposure.
- **Strategic AI Technical Lead — Mandela Legacy Foundation:** Specifically outsourced to the Mandela Legacy Foundation as AI and Cloud Technical Lead following high-performance delivery at Netcampus — selected for advanced Azure AI Language and cloud infrastructure proficiency to bridge legacy data with modern AI-driven automation. Owned architectural decisions and led the digital transformation programme end-to-end.
- **DevOps & Observability:** Managed complex cloud environments and CI/CD pipelines; implemented automated monitoring, alerting, and testing protocols ensuring high availability for mission-critical AI-powered cloud applications.

Software Developer

Cape Code Collective | South Africa *January 2024 – February 2025*

- **Full-Stack Development:** Delivered customised full-stack web solutions leveraging Python, HTML5, CSS3, JavaScript, PHP, React.js, and TypeScript — building high-quality, responsive, client-facing applications with emphasis on performance and reliability.
- **API Integration & Data Exchange:** Developed and integrated RESTful APIs to streamline platform data exchange, enhance interoperability, and lay groundwork for future automated and AI-augmented data workflows.
- **Agile Delivery & Client Management:** Applied sprint-based Agile practices to design and deploy key features iteratively; worked directly with clients to translate requirements into technical specifications and ensure delivery satisfaction.

Systems Trainee

JMM Mining Pty Ltd | Johannesburg, South Africa *January 2022 – November 2023*

- **Operations Intelligence & Data Systems:** Designed and deployed internal data dashboards and tracking tools for operational data visualisation and mining data analysis, enabling leadership to transition to proactive, data-driven decision-making.
- **Cloud Migration & Infrastructure:** Facilitated the organisation's migration to secure cloud storage and email systems, implementing reliable networking solutions and establishing foundational data governance principles.

EDUCATION

Diploma in Software Development | Rosebank IIE | Cape Town, South Africa

Feb 2023 – Nov 2025

- **Advanced Data Structures & Integrity:** Developed core expertise in relational database design, focusing on Entity-Relationship Diagrams (ERD), 3NF Normalisation, and SQL DDL/DML for maintaining high data integrity and structure.
- **Quantitative Modelling & Analytics:** Applied mathematical and statistical methods to perform rigorous data analysis, demonstrating ability to express real-world problems in mathematical terms and derive reliable quantitative solutions.
- **System Intelligence Foundation:** Gained foundational knowledge leveraging Information Systems (TPS/MIS) to generate actionable business intelligence and researched the strategic importance of Big Data (Volume, Variety, Velocity) and Data Warehousing/Marts.

Azure Web Services Training | Netcampus (Kaagisano Group) | Cape Town, South Africa

Feb 2025 – Feb 2026

AWS Practitioner Exam Preparation | Amazon Web Services | Cape Town, South Africa

Jun 2023 – Feb 2025

PROFESSIONAL CERTIFICATIONS

2025: Microsoft Certified: Azure AI Fundamentals

2025: Rosebank IIE — Diploma in Information Technology and Software Development

2024: AWS Certified Cloud Practitioner

2025: Azure Data Engineering Certification (In Progress)

2025: Google Digital Marketing & E-commerce Certifications

2022: Coursera — Introduction to CRM with HubSpot

2022: Google — Foundations of Digital Marketing and E-commerce

NON-ACADEMIC INTERESTS & ACTIVITIES

- **Creative Content Contributor – COREACTIVITIES®:** Produced and shared engaging content on arts, culture, and videography, growing the brand audience to 465 followers across 60+ posts. Curated weekly playlists and visually compelling videography to enhance brand identity and audience loyalty.
- **Basketball Enthusiast:** Active player — quick thinking, discipline, and high-performance teamwork under pressure.
- **Strategic Thinking:** Maintained a competitive daily rating on Chess.com, reflecting analytical depth and forward-planning capability.
- **Artistic Exploration:** Passionate about arts and culture, continually engaging with emerging creative trends to inspire innovative thinking and multidisciplinary projects.

PROFESSIONAL REFERENCES

Tate Modimogale

Managing Director, JMM Mining Pty Ltd

tate@kmining.co.za | +27 79 877 9652

Ms. Maditi

Netcampus Group

(021) 831 0550 (Cape Town Office)